

### **Question 1: Economic Data [35 Points]**

**This question will ask you to get familiar with economic data, plotting it, and drawing conclusions.**

**Please use the Federal Reserve Economic Data (FRED) database, maintained by the Federal Reserve Bank of St. Louis (link here). For your reference, note that the National Bureau of Economic Research (NBER) is the entity that dates the business cycles in the US (link here). Please be precise about what units you are using and don't forget to include a title and axis labels in all charts, you can use any software you'd like, e.g. Excel is fine.**

**Question 1.1 [10 Points]** Download and plot the series of US Real Gross Domestic Product (code GDPC1, billions of chained 2012 dollars annual rate) at a quarterly frequency, from the earliest available date to the last available date. What is the fourth quarter of 2021 real GDP? At a quarterly frequency, how much did real GDP fall “from peak to trough” during the COVID-19 recession? (i.e.: from the maximum RGDP before the recession, to the lowest point during the recession.) How much did real GDP fall from peak to trough during the Great Recession (2007-2009)?

Q1.1.1 What is the fourth quarter of 2021 real GDP?

Answer: 21082.134B

Q1.1.2 At a quarterly frequency, how much did real GDP fall “from peak to trough” during the COVID-19 recession?

Answer: 1907.456B

Q1.1.3 How much did real GDP fall from peak to trough during the Great Recession (2007-2009)?

Answer: 646.045999999998B

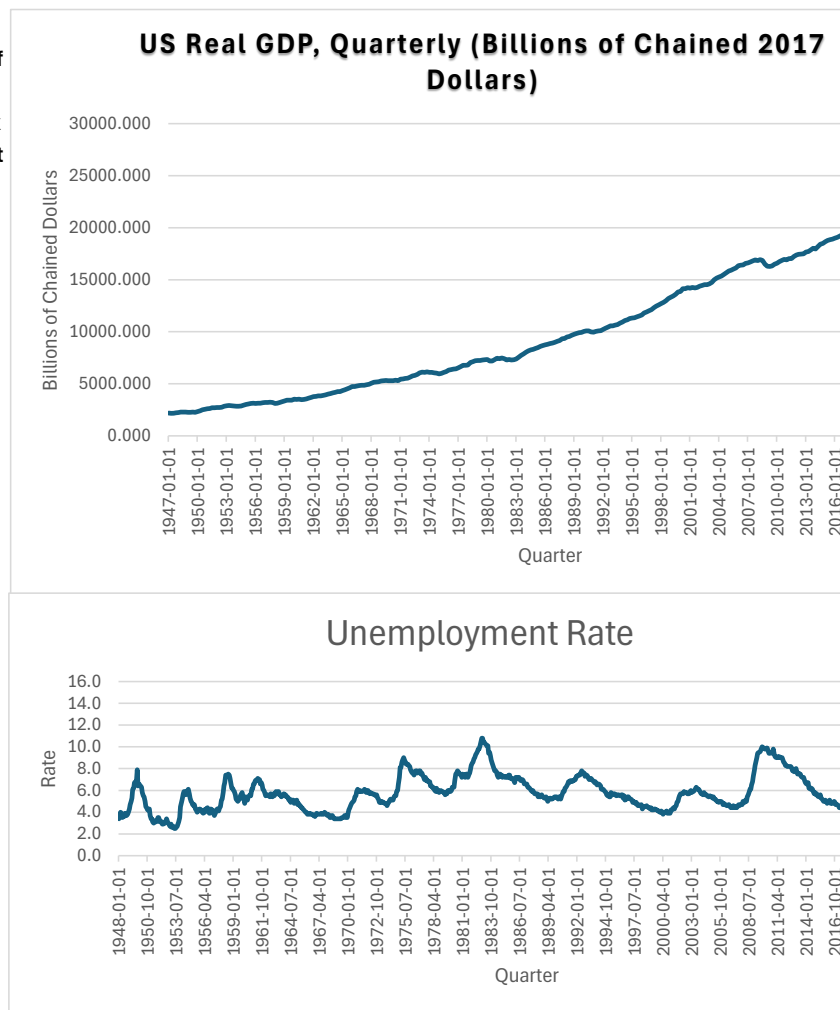
**Question 1.2 [5 Points]** Download and plot the series for quarterly unemployment (UNRATE), from the earliest available date to the latest. How does the unemployment rate in Q4 2019 compare to Q2 2020? How do the changes in unemployment rate now compare to the period between Q2 2008 and Q2 2009?

Q1.2.1 How does the unemployment rate in Q4 2019 compare to Q2 2020?

Answer: The UR in Q4 2019 was 3.5 while Q2 2020 was 11, a huge crash due to the Covid recession

Q1.2.2 How do the changes in unemployment rate now compare to the period between Q2 2008 and Q2 2009?

Answer: As of May 2026 The given report is 4.3 why the AVG between Q2 2008-Q2 2009 is 6.8



**[10 Points]** Using the real GDP series you downloaded in (1), compute the quarterly output growth  $GDPT - GDPT-1$  rate as  $gt = \frac{GDPT - GDPT-1}{GDPT-1} \times 100$ . Next, using the unemployment rate series you downloaded in (2),  $GDPT-1$  compute the quarterly change in the unemployment rate as  $\Delta u = u_t - u_{t-1}$ . Plot  $\Delta u$  on the y-axis, and  $gt$  on the x-axis over the period 1960-2019, and over the period 1960-2022. Describe the pattern you find in words. Are the changes in output and unemployment during the 2008 financial crisis and the COVID-19 recession qualitatively consistent with what we would expect from Okun's law?

Q1.3.1 compute the quarterly change in the unemployment rate as  $\Delta u = u_t - u_{t-1}$ . Plot  $\Delta u$  on the y-axis, and  $gt$  on the x-axis over the period 1960-2019, and over the period 1960-2022. Describe the pattern you find in words  
 Answer: what we see here is called the Okun's law which shortly put says: "for every 1% a nations UR decreases said nations GDP growth goes up by 2%"

Q1.3.2 Are the changes in output and unemployment during the 2008 financial crisis and the COVID-19 recession qualitatively consistent with what we would expect from Okun's law?  
 Answer: Yes exactly, because UR went up and USA's GDP dropped

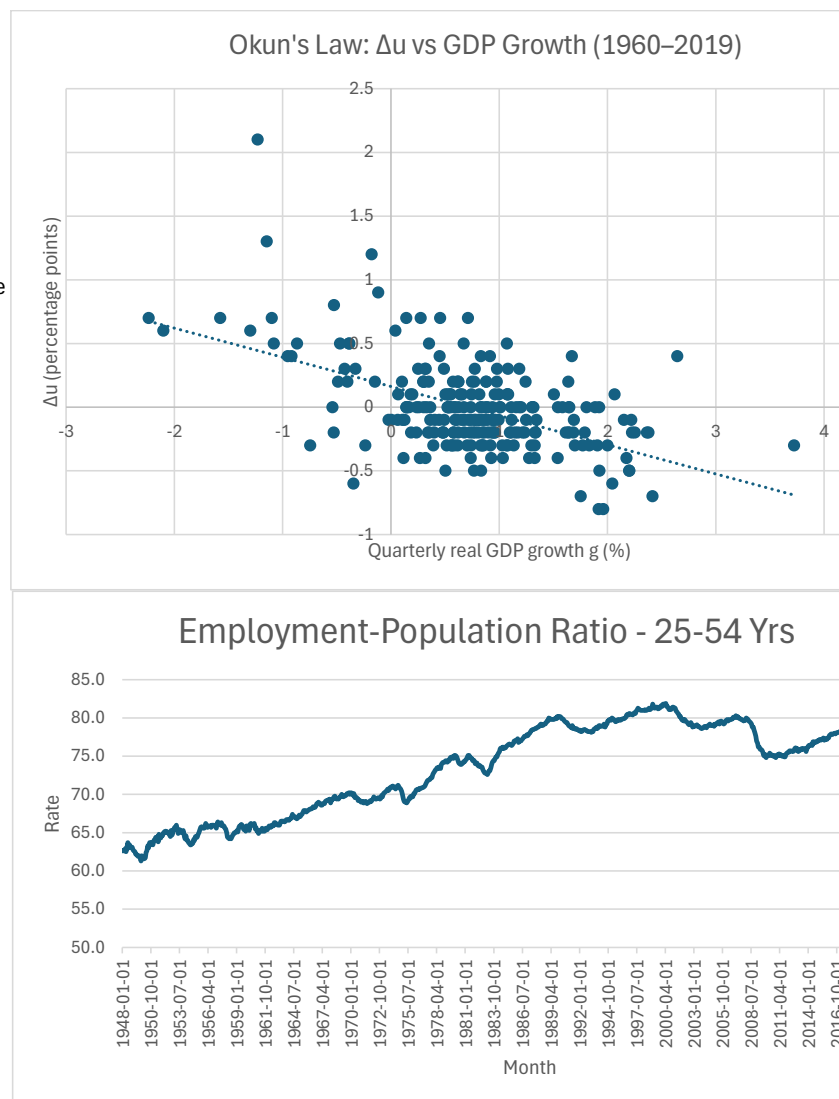
**[5 Points]** Download the quarterly Employment to Population Ratio for 25-54 year olds (LNS12300060), from the earliest available date to the latest. Plot the non-employment to population ratio, calculated as  $100\% - \text{employment to population ratio}$ . How does the non-employment to population ratio in Q4 2019 compare to Q2 2020? How do the changes in the non-employment to population ratio compare to the period between Q2 2008 and Q2 2009?

Q1.4.1 How does the non-employment to population ratio in Q4 2019 compare to Q2 2020?  
 Answer: The non-employment to population ratio in said dates in order are 19.7% and 30.4% which is a huge rise again due to the pandemic and the scattering of the jobs market

Q1.4.2 How do the changes in the non-employment to population ratio compare to the period between Q2 2008 and Q2 2009?  
 Answer: The non-employment to population ratio in said dates in order are 20.5% and 24% which is again a huge rise due to the financial crisis of 2008 however not as much of a dramatic crash contrary to the covid crash but this can be explained by Corsetti's laws

**[5 Points]** Compare the relationship between the evolution of the Unemployment Rate and the evolution of the Non-Employment to Population Ratio from peak to trough of the COVID-19 recession and the 2008 financial crisis. Why is there not a 1-1 relationship between the two variables (write down the formulas for each one)? What can you conclude about labor force participation during the COVID-19 recession (feel free to use FRED to answer this question if helpful)?

Q1.5.1 Why is there not a 1-1 relationship between the two variables (write down the formulas for each one)?



Answer: so firstly the said formulas are first  $\text{Unemployment Rate} = \text{Labor Force} / \text{Unemployed while Non-Employment to Pop. Ratio} = \text{Total Population} / \text{Not Employed}$  so if we look at only the unemployment rate when trying to grasp the damage done by said crises it down plays how much of the people lost jobs, when we solely look at the unemployment rate we look at the ratio of the people who are in said job market but aren't finding a job but when we look at real life we saw a lot of people lose their jobs and simply stop looking for one, why? well simply because of said crises people rather stay home, look after their family, protect themselves (only applicable to covid). so that's why when we look at these two different formulas we might get different ideas because just because someone isn't looking for a job it doesn't mean they weren't forced out of one.

Q1.5.2 What can you conclude about labor force participation during the COVID-19 recession (feel free to use FRED to answer this question if helpful)?

Answer: what we can conclude from the point being made is that, during covid a lot of people left their jobs and continued looking for one, but even more people left them and gave up, hence why these two variables are skewed

### **Question 2: Measuring nominal and real GDP [30 Points]**

**Consider the following closed-economy in which there are only three goods: masks, movies, and air travel. The following table gives the prices and quantities consumed of the three goods for 2019 and 2020, and 2021:**

|           | 2019  | 2019     | 2020   | 2020      | 2021   | 2021      |
|-----------|-------|----------|--------|-----------|--------|-----------|
| Goods     | Price | Quantity | Price2 | Quantity3 | Price4 | Quantity5 |
| Movies    | 0.1   | 60       | 2      | 400       | 0.5    | 350       |
| AirTravel | 5     | 90       | 5      | 10        | 5      | 60        |
| Masks     | 100   | 15       | 40     | 10        | 75     | 20        |

**[5 Points] Compute the Nominal GDP (NGDP) of each year, show your work.**

Answer: So to compute our NGDP we are going to do  $Q \times P$  for each good and sum them.

2019 NGDP = 1956

2020 NGDP = 1250

2021 NGDP = 1975

**[5 Points] Compute the real GDP (RGDP) for each year using 2019 as the base year. Then compute annual RGDP growth for each year and express it as a percentage.**

Answer: so first let's calculate the RGDP for each year taking 2019 as the base

2019 RGDP = 1956

2020 RGDP = 1090

2021 RGDP = 2335

Now to express it as a percentage we are gonna do  $((\text{GDP} - \text{GDP}^*) / \text{GDP}^*) \times 100$

2020: -44.2740286298569

2021: 114.220183486239

**[10 Points] Compute the real GDP (RGDP) for each year using 2020 as the base year. Then compute RGDP growth and express it as a percentage. Explain why it is different from the real GDP growth computed using 2019 as the base year. Do the same using 2021 as the base year.**

Answer: so first let's calculate the RGDP for each year taking 2021 as the base

2019 RGDP = 1605

2020 RGDP = 1000

2021 RGDP= 1975

Now to express it as a percentage we are gonna do  $((GDP - GDP^*) / GDP^*) \times 100$

2020: -37.6947040498442

2021: 97.5

So the main question, why is it that these are different? Well the reason is that we are using different years prices, so for example in 2021 masks were cheaper in relative to movies and movies were more expensive in relative to 2019 so essentially movies had more weight and masks had less weight relative to 2019.

**[10 Points] The BEA has adopted a different method to compute growth rates (it is explained in full in the appendix of Chapter 2). The formula is given by:**

$g_{T-t, T} = \frac{g_{T-t, T} + g_{T-t, T}}{2}$  BEA  $g_{T-t, T} = 2$

Where BEA  $g_{T-t, T}$  is the growth rate calculated by the BEA between years T and T - t. And  $g_{T-t, T(b)}$  is the growth rate between year T and T - t using b  $\in \{T, T - t\}$  as the base year. Use this method to calculate the growth rate between 2021 and 2020, and between 2020 and 2019. Is this a better method? Explain.

Answer: so let's use the formula given, firstly let's do 2020base - 2021

after doing so we get the average: 105.860091743119

Now let's do 2019base - 2020

After doing so we get the average: -40.9843663398505

This does in fact give us a better method because now we are averaging goods prices/weights over the two years as well

### **Question 3: The Goods Market [35 Points]**

**Consider the following economy. The demand for goods, Z, is given by**

**$Z \equiv C + I + G$  (1)**

**where C is consumption, I is investment, and G is government spending. The consumption function is given by**

**$C = c_0 + c_1 YD$  (2)**

**where  $c_0 \geq 0$ ,  $0 < c_1 < 1$ ,  $YD \equiv Y - T$  is disposable income, and T is taxes.**

**When we say "exogenous variables", this refers to variables determined outside of the model such as, for example  $c_0$ ,  $c_1$ . In contrast, C, Y are endogenous variables. Hereafter, we assume T is either exogenous, or endogenous as specified below, while G and I will always be treated as exogenous**

**[5 Points] Describe in words what  $c_0$  and  $c_1$  represent.**

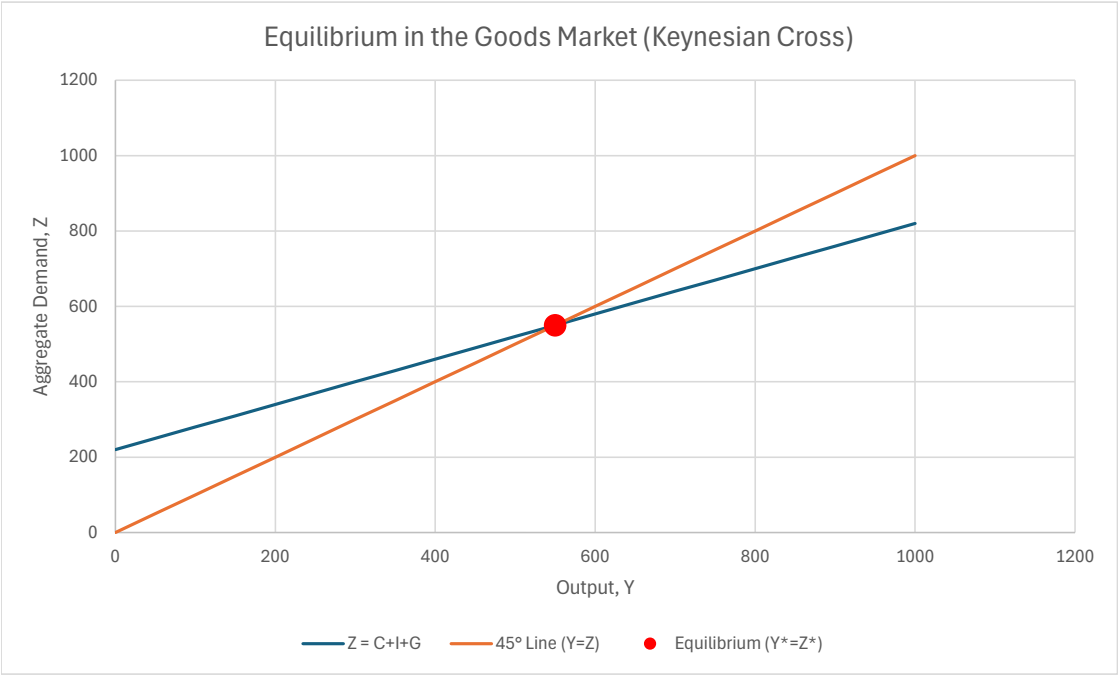
Answer:  $c_0$  here is the average spend on a singular dollar, say we give 1000 people a dollar each and the average spent by each person was 50c then  $c_1$  would be 0.5,  $c_0$  on the other hand is autonomous spending, such as consumer confidence

**[5 Points] What equilibrium condition can you use to solve for equilibrium output? Explain it.**

Answer: the EQL condition here would be  $Z=Y$  , were  $z$  is AGGR demand and  $y$  is output

[5 Points] Draw a graph representing equilibrium in the goods market, appropriately label all axes.

Answer: The equilibrium condition is  $Z=Y$ . Below is a graph of the aggregate demand line  $Z = c_0 + c_1(Y-T) + I + G$  against the 45° line ( $Z=Y$ ). Their intersection is the equilibrium output.



**Assumptions (illustrative example values)**

|                 |     |
|-----------------|-----|
| $c_0$ (autonomi | 100 |
| $c_1$ (margin:  | 0.6 |
| $I$ (investme   | 50  |
| $G$ (governm    | 100 |
| $T$ (taxes)     | 50  |
| Equilibrium     | 550 |
| Equilibrium     | 550 |

| Y (Output) | Z = C+I+G | 45° Line (Y=Z) |
|------------|-----------|----------------|
| 0          | 220       | 0              |
| 50         | 250       | 50             |
| 100        | 280       | 100            |
| 150        | 310       | 150            |
| 200        | 340       | 200            |
| 250        | 370       | 250            |
| 300        | 400       | 300            |
| 350        | 430       | 350            |
| 400        | 460       | 400            |
| 450        | 490       | 450            |
| 500        | 520       | 500            |
| 550        | 550       | 550            |
| 600        | 580       | 600            |
| 650        | 610       | 650            |
| 700        | 640       | 700            |
| 750        | 670       | 750            |
| 800        | 700       | 800            |
| 850        | 730       | 850            |
| 900        | 760       | 900            |
| 950        | 790       | 950            |
| 1000       | 820       | 1000           |

[5 Points] Solve for equilibrium output  $Y$  assuming that  $G$ ,  $T$  and  $I$  are exogenous.

Answer: Assuming so will give us  $Y=C+I+G$

[5 Points] What is the autonomous spending ( $c_0$ ) multiplier in this case? That is, how much does equilibrium output change in response to an increase in  $c_0$ ?

Answer: it would be  $1/(1-c_1)$

[5 Points] Let  $S \equiv Y - T - C$  denote the private saving. Express  $S$  in terms of  $I$ ,  $T$ , and  $G$ .

Answer: so lets start with  $Z= C + I + G$ , then we can solve for  $C=Y - I - G$  and get  $S= C + I +G - T - Y + I + G$  and finally we can get  $S = C + 2I +2G - T - C - I - G$  and there for  $S = I + G - T$

[5 Points] How much does private saving  $S$  increase in response to an increase in  $G$ ? (Use your answer to part 3)

Answer: Assuming a S of I + G - T it would be directly correlated , 1 of 1

**SELF-STUDY EVALUATION — MIT 14.02 (OCW) Problem Set 1**

|                                              |           |
|----------------------------------------------|-----------|
| Q1: Economic Data (out of 35)                | 26        |
| Q2: Measuring Nominal & Real GDP (out of 30) | 19        |
| Q3: The Goods Market (out of 35)             | 27        |
| <b>TOTAL (out of 100)</b>                    | <b>72</b> |

Letter grade **C**

Grader notes: Strong computational work (all Q2.1–2.2 figures independently verified correct; solid 45°-line equilibrium graph; correct multiplier and  $S = I + G - T$ ). Deductions for: Q2 2020 unemployment misread  
*Graded independently against the official MIT 14.02 PS1 solution key, with all arithmetic re-verified from the student's own data. — Claude (Anthropic), AI grading assistant in Excel, July 2, 2026*